

Salman Haider

GIS Specialist

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SUMMARY

GIS professional with 4+ years of applied experience across Canada, Australia, Pakistan, and international client markets, supported by a completed MGIS from the University of Calgary. Combines production GIS, spatial analysis, ArcGIS Enterprise, remote sensing, spatial databases, Python/ArcPy automation, and client-facing delivery, including 105+ franchise territory projects.

CORE QUALIFICATIONS

- Delivered end-to-end GIS workflows from requirements and data preparation through analysis, QA/QC, mapping, and reporting.
- Automated repetitive geospatial workflows using Python, ArcPy, SQL, APIs, OpenCV, and reusable GIS tools.
- Built interactive GIS products using ArcGIS Online, Experience Builder, Dashboards, Survey123, Leaflet, and web technologies.
- Translated technical analysis into practical maps, applications, reports, and recommendations for clients, policy staff, and non-GIS users.

EXPERIENCE

GIS Research Assistant

University of Calgary

📅 01/2026 - 04/2026 📍 Calgary, Canada

- Analyzed access to fresh-food destinations across Calgary using scheduled public transit, walking, population geography, street networks, GTFS data, and destination datasets.
- Maintained City of Calgary street-centreline geometry and attributes within an enterprise geodatabase using ArcGIS Pro, editing versions, and structured quality checks.
- Published and updated GIS layers through ArcGIS Enterprise Portal after reviewing schemas, spatial references, fields, naming, symbology, and sharing requirements.
- Built a multimodal ArcGIS Network Analyst workflow and identified 33,212 residents in dissemination areas classified with no or poor fresh-food access.
- Developed an ArcGIS Experience Builder application allowing non-GIS users to locate areas and explore accessibility classes, travel times, and reachable destinations.

Volunteer Student Research Assistant

BC Ministry of Water, Land and Resource Stewardship

📅 05/2025 - 11/2025 📍 Remote, Canada

- Supported economists and policy analysts with GIS maps, charts, graphs, and spatial evidence for forestry, protected-area, and greenhouse-gas analysis.
- Analyzed forest coverage inside and outside Legal and Non-Legal Old Growth Management Areas and TANTALIS protected areas using provincial VRI datasets.
- Applied overlay, intersection, spatial joins, geometry calculations, and summary statistics using Python and R to produce area totals, percentages, and comparative graphics.
- Checked coordinate alignment, geometry, overlap, completeness, and potential double counting before preparing maps and visualizations for internal briefings.

GIS Analyst (Franchise Territory Mapping)

Franchise Ready

📅 11/2021 - 08/2024 📍 Remote, Australia

- Delivered end-to-end location-intelligence and territory-design services for 105+ franchise projects across Australia and additional international client markets.
- Translated customer profiles, demographics, purchasing power, competitors, service ranges, locations, sales patterns, and expansion goals into measurable spatial criteria.
- Built and maintained reusable census, business, demographic, land-use, customer, competitor, and point-of-interest datasets for recurring commercial analysis.
- Developed a multi-source business-location database in PostgreSQL/PostGIS using Python-based collection, cleaning, deduplication, categorization, geocoding, and reconciliation workflows.
- Produced territory systems, scoring models, Excel reports, maps, ArcGIS Online content, dashboards, and branded Mango Maps products for client decision support.
- Evaluated web-GIS platforms, recommended Mango Maps, supported vendor engagement, administered the organizational account, managed user access, and onboarded clients.
- Created Python and ArcPy automation for data processing, geocoding, reporting, spatial analysis, map production, quality checks, and repeatable production workflows.

EXPERIENCE

GIS Associate (Land Record Digitization)

Greenage Services

📅 04/2021 - 11/2021 📍 Islamabad, Pakistan

- Supported a government state-land digitization project converting historic cadastral paper and cloth records into structured parcel and state-land GIS datasets.
- Collected RTK GPS ground control, supported drone and satellite imagery rectification, georeferenced cadastral sheets, and digitized parcels and related land features.
- Integrated scanned maps, orthophotos, RTK observations, CAD drawings, and existing GIS layers, including CAD-to-GIS conversion using FME Workbench.
- Built a Python, OpenCV, and OCR workflow that automatically sorted thousands of scanned cadastral sheets by locality, removing a major production bottleneck.
- Developed ArcPy reporting and QA/QC routines for production monitoring, unreadable records, missing attributes, inconsistent values, and data-quality exceptions.

GIS Survey Technician

MER Solutions

📅 10/2020 - 04/2021 📍 Rawalpindi, Pakistan

- Supported the Benazir Income Support Programme household survey through boundary mapping, Survey123 development, household interviews, spatial linkage, and data validation.
- Georeferenced government maps and produced field layers showing survey boundaries, roads, settlements, dwellings, landmarks, access routes, and household locations.
- Designed and maintained ArcGIS Survey123 forms for demographic, socio-economic, property, asset, and land information collected during household interviews.
- Revised form questions, field options, validation rules, and structure using direct field observations and feedback from survey teams.
- Linked Survey123 records with mapped household locations and supported verification, confidentiality-sensitive data handling, validation, and final reporting.

EDUCATION

Master of Geographic Information Systems (MGIS)

University of Calgary

📅 08/2024 - 05/2026 📍 Calgary, Canada

MS Environmental Science – Graduate Coursework

Bahria University

📅 03/2023 - 12/2023 📍 Islamabad, Pakistan

- Coursework included Environmental Management, Climate Change Adaptation and Mitigation, Environmental Analytical Techniques, Disaster Management, Health Safety and Environment, Solid & Hazardous Waste Management, and Environmental Risk Assessment

Bachelor of Science - Geoinformatics and Earth Observation

Arid Agriculture University

📅 09/2016 - 10/2020 📍 Rawalpindi, Pakistan

TECHNICAL TOOLS

GIS Tools

ArcGIS Pro, ArcGIS Enterprise and Portal, enterprise geodatabases, ArcGIS Online, ArcGIS Dashboards, ArcGIS Experience Builder, Survey123, ArcGIS Network Analyst, ArcGIS Business Analyst, Spatial Analyst, ModelBuilder, ArcMap, QGIS, Google Earth Engine, Mango Maps, Leaflet, Mapbox

Data & Development

Python, ArcPy, SQL, JavaScript, R, HTML, CSS, pandas, NumPy, GeoPandas, GDAL, Shapely, Flask, FastAPI, Docker, Git, GitHub

Spatial Databases and Data Engineering

PostgreSQL, PostGIS, pgRouting, enterprise geodatabases, spatial SQL, schema design, ETL, geocoding, data normalization, deduplication, spatial indexing

Other

Microsoft Excel, PowerPoint, Word, Charts, Graphs, Technical reports

SELECTED PROJECTS

ArcPy Forestry Sampling-Point Generator Tool

- Built a reusable ArcGIS Pro tool to identify suitable forestry sampling areas and automatically generate field-sampling points from operational criteria.
- Applied forest-composition, slope below 5°, and road-distance within 1,000 metres criteria to construct candidate sampling areas.
- Validated NAD 1983 10TM AEP Forest coordinates and automated buffering, clipping, slope analysis, raster-vector conversion, and random-point generation with ArcPy.

LiDAR and Object-Based Conifer Tree-Height Classification

- Analyzed conifer forest structure using a 1-metre LiDAR Canopy Height Model, QuickBird imagery, and object-based classification techniques.
- Classified conifers into 9 five-metre height classes from 0–45 m; the largest 25–30 m class covered 307,565 m².
- Compared KNN and SVM image classifiers, selected SVM for stronger small-patch performance, and documented classification confusion and independent-validation requirements.

Spatio-Temporal Conflict Pattern Analysis and Severity Prediction Using ACLED Data

- Analyzed 109,337 conflict events across Syria, Iraq, and Palestine from 2022–2024 through temporal trends, mapping, administrative aggregation, KDE, and severity analysis.
- Built rare-event logistic regression and decision-tree models, using class balancing, cross-validation, threshold tuning, precision, recall, F1, and limitation-aware interpretation.

Geo Copilot – Natural-Language GIS Tool for ArcGIS Pro

- Built an ArcGIS Pro script tool that converts plain-language GIS requests into controlled geoprocessing commands using Large language Models and layer metadata and an explicit tool allow-list.
- Evaluated 36 test scenarios, achieving 91.67% function accuracy and 83.33% parameter accuracy with the GPT-based workflow.

Logistics Routing Application

- Built an end-to-end routing application from raw OpenStreetMap data through network topology, routing costs, one-way restrictions, route alternatives, and interactive mapping.
- Developed Flask APIs and a Leaflet interface supporting origin/destination routing, blocked polygons, traffic-delay points, distance or time optimization, and alternatives.

Sentinel-1 SAR Ship-Detection Pipeline

- Built a pipeline that tiled full Sentinel-1 VV scenes into 640 × 640 patches while preserving offsets for reconstruction into source-scene coordinates.
- Processed scenes containing over 2,500 patches, evaluated detection performance using precision, recall, and F1 score, and applied water-based filtering to reduce land false positives.

Namal Lake Desiccation Assessment, 2000–2024

- Assessed 25 years of water-area change, agricultural expansion, and climate variables using MNDWI, NDVI, annual imagery composites, and ERA5-Land data.
- Built a standardized OLS model with adjusted R² = 0.594, identifying agricultural area as the strongest negative predictor of lake area.

Ord's Kangaroo Rat Habitat-Suitability Model

- Built a repeatable habitat-suitability workflow using DEM, active sand dunes, roads, fireguards, ecological range sites, species occurrences, aspect, and proximity.
- Combined primary and secondary habitat surfaces using documented 70% and 30% weightings to support ecological suitability assessment.

ArcGIS Drone Route and Report Automation

- Created an ArcGIS Python toolbox comparing driving routes with straight-line drone routes using distance, travel time, payload, range, and feasibility criteria.
- Automated route layers, operational summaries, time-savings calculations, greenhouse-gas savings, Excel-based inputs, and GIS reporting.

ADDITIONAL INFORMATION

- Valid Canadian open work permit
- Valid BC Class 4 commercial driver's licence
- Canadian Red Cross Standard First Aid and CPR/AED Level C
- Available to relocate for the role
- References available upon request